

Challenge Write-Up Template

Challenge Information

Student Name	Daniella Marie A. Leonud
Section / Class	BSIT-3G
Challenge Name	MultiCode
Category	General Skills
Points / Difficulty	Easy
Date Solved	August 8, 2026
Flag Format Given	picoCTF{nested_enc0ding_0c2192d7}

1. Challenge Description

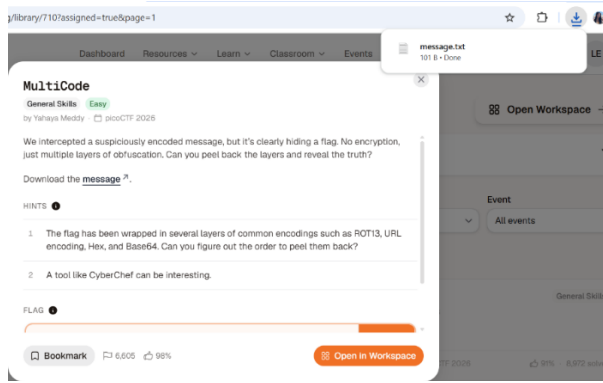
Restate the challenge in your own words: what were you given, and what were you asked to find or do? Do not simply copy-paste the original prompt.

In this MultiCode challenge, I was given an encoded message inside a text file named **message.txt**. My task was to decode the message and find the hidden flag. When I read the challenge hints, I noticed that CyberChef was mentioned. The hint also gave clues about the different encoding layers that were used. From this, I understood that the message was not going to be decoded using only one method. I needed to remove several layers of encoding in the correct order until the actual flag appeared. The different layers involved were Base64, Hex, URL encoding, and ROT13.

2. Initial Analysis / Reconnaissance

What did you check first? What did file, strings, a webpage view-source, or another first look tell you? What category did you decide this was, and why?

First, I opened the challenge and carefully read the instructions and hints. I noticed that the challenge provided a **message.txt** file, so I decided to download it and check what was inside. After opening the file, I saw that the content was not readable as normal text. Instead, it was a long encoded string. This told me that the main goal of the challenge was probably to decode the string rather than trying to find a vulnerability in a program. The hint was very helpful because it mentioned CyberChef and gave clues about the encoding layers. Because of this, I decided to use CyberChef instead of trying to manually decode every layer.



3. Tools and Commands Used

List every tool/command you used, in the order you used them, and what each one told you.

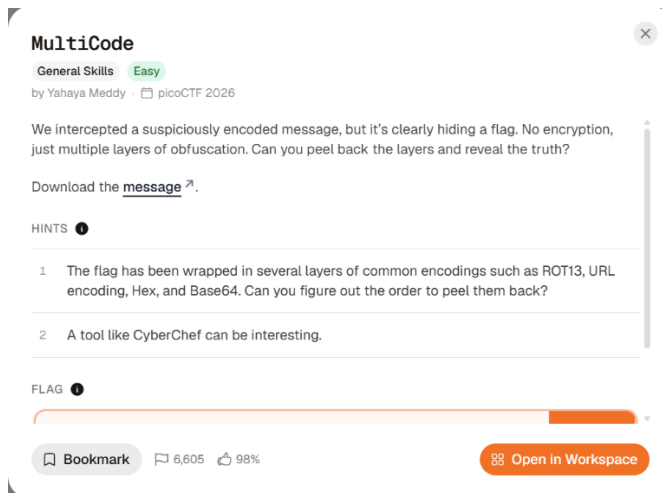
Tool / Command	What it was used for	Result / Output
Browser (Google Chrome)	Downloaded and opened message.txt to view the encoded text.	Revealed the Base64 input string.
CyberChef	Used to build a sequential decoding recipe (From Base64, From Hex, URL Decode, ROT13).	Automatically decoded all layers and revealed the final flag.

4. Solution Walkthrough

Explain your reasoning step by step — not just what you typed, but why you tried it and what you concluded from the result. Include screenshots, code snippets, or command output where useful. A reader who has not seen the challenge should be able to follow your logic and arrive at the same answer.

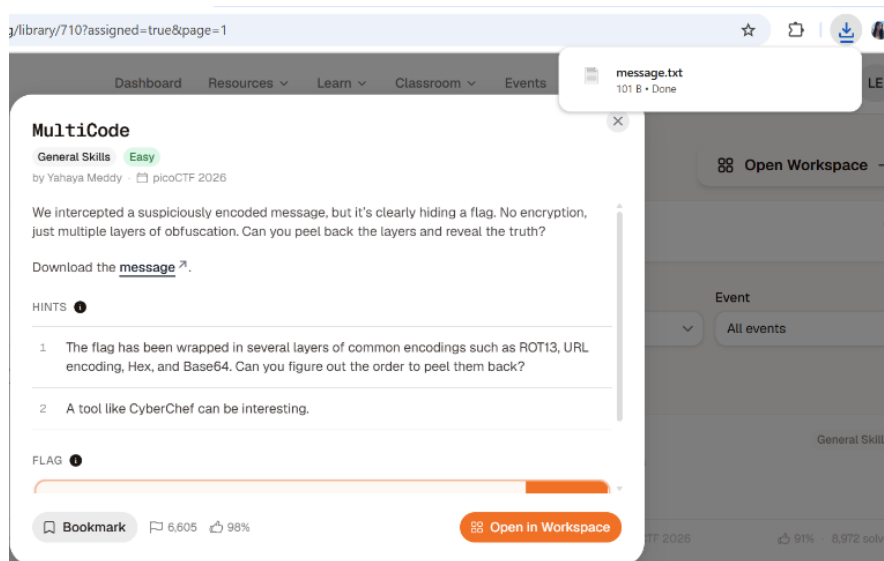
Step 1

First, I opened the MultiCode challenge and read the instructions and hints. The hint was important because it mentioned **CyberChef** and gave clues about the different encoding layers. From the information given, I understood that I needed to decode the message several times. The layers mentioned were **From Base64, From Hex, URL Decode** and **ROT13**. I also understood that the order was important. If I used the operations in the wrong order, the encoded message would not decode correctly.



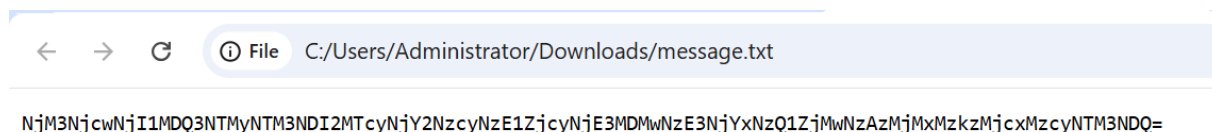
Step 2

I clicked the **message.txt** link provided in the challenge and downloaded the file. After downloading it, I opened the file and looked at its contents. Instead of normal readable text, I saw an encoded string. This confirmed that the message needed to be decoded before I could find the flag.



Step 3

I copied the entire encoded string from **message.txt**. Then I opened the CyberChef website because the challenge hint specifically suggested using it. CyberChef is useful for this type of challenge because it allows multiple decoding operations to be placed together in a recipe.

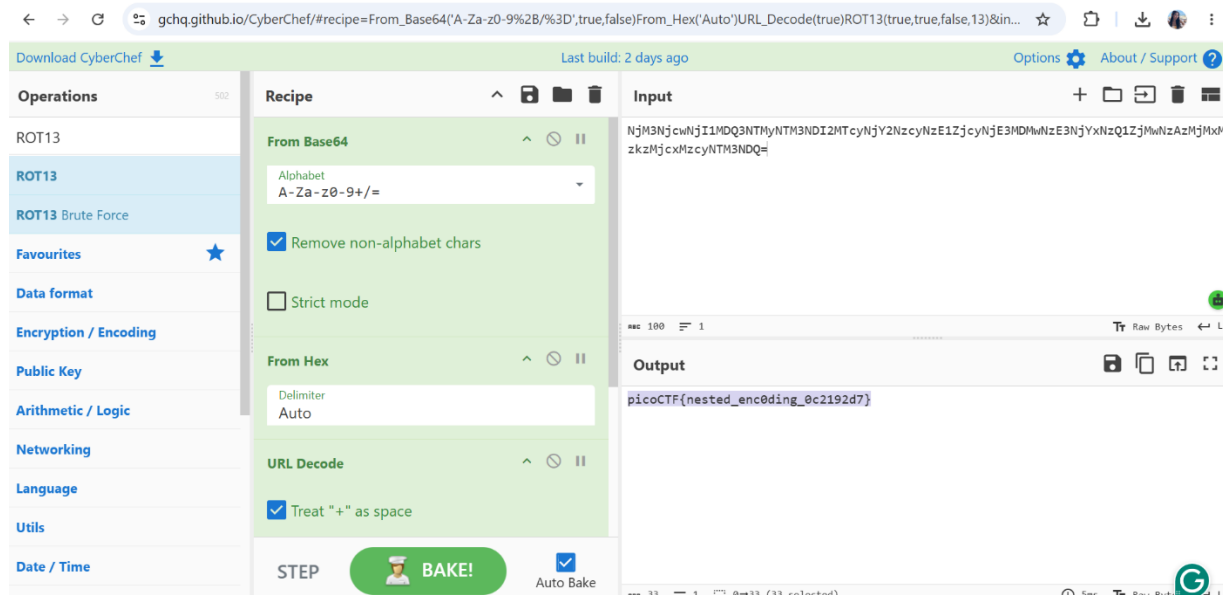


Step 4

After opening CyberChef, I pasted the encoded string into the Input box. I dragged the following operations into the recipe:

1. From Base64
2. From Hex
3. URL Decode
4. ROT13

I placed them in this exact order because the message had been encoded using several layers.



After adding all four operations, CyberChef automatically processed the input. The first operation removed the Base64 layer. The next operation decoded the Hex representation. After that, the URL encoding was removed, and finally, ROT13 was applied to reveal the readable text. The output box then displayed the actual flag **picoCTF{nested_encoding_0c2192d7}**

Step 5

After seeing the flag in the output box, I copied it and went back to the CyLab challenge page. I pasted the flag into the submission box and submitted it. The challenge accepted the flag, which confirmed that I had successfully completed the MultiCode challenge.



5. Flag

```
picoCTF{nested_enc0ding_0c2192d7}
```

6. Key Concept / What I Learned

In 2-4 sentences, name the underlying security concept or vulnerability class this challenge taught (e.g. "weak RSA exponent," "reflected XSS," "stack buffer overflow"). This is the part of the write-up most worth remembering after the flag is submitted.

This challenge taught me how data can be hidden under many layers like Base64, Hex, URL encoding, and ROT13. I learned how to use CyberChef to remove each layer step-by-step. I realized that reading the hints carefully makes solving the problem much faster.

7. References

List any external resource you consulted beyond course materials (a tool's documentation, a technique explainer, a similar public write-up). It is normal and expected to use references — just cite them.

CyberChef Web Utility (<https://gchq.github.io/CyberChef/>)
